

Fraud Detection Analytics using SQL Server

Internship Project Documentation

Submitted By -

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Project Title: Fraud Detection Analytics using SQL Server

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1. Introduction

Fraud detection is one of the most important applications of data analytics in the banking and financial sector. Every day, millions of digital transactions are processed through online banking systems, mobile payment applications, and financial institutions.

Detecting fraudulent transactions manually is difficult because of the huge volume of data.

This project uses Microsoft SQL Server to analyze financial transaction records and identify suspicious transaction patterns.

Advanced SQL techniques such as Common Table Expressions (CTEs), Recursive CTEs, Window Functions, Aggregate Functions, and Joins are applied to detect fraud, analyze high-risk accounts, validate balances, and generate business insights that can support fraud prevention strategies.

2. Project Objectives

The objectives of this project are:

- Detect fraudulent financial transactions.
- Identify suspicious customer accounts.
- Analyze high-value money transfers.
- Validate transaction balances.
- Detect recurring fraud patterns.
- Generate business insights for fraud prevention.
- Demonstrate advanced SQL analytical techniques.

3. Business Problem

Financial institutions process millions of transactions every day. Due to the enormous volume of data, manually identifying fraudulent transactions is both time-consuming and inefficient. Fraudulent activities such as unauthorized money transfers, suspicious account behavior, and abnormal balance changes can lead to significant financial losses if not detected in time.

This project aims to analyze transaction data using SQL Server to identify suspicious activities, detect fraudulent transaction patterns, validate account balances, and provide actionable insights. The analysis helps organizations improve fraud monitoring, strengthen security measures, and support data-driven decision-making.

4. Dataset Description

The project uses the **PaySim Mobile Money Transactions Dataset**, which simulates real-world financial transaction data.

Dataset Information

- Dataset Name: PaySim Mobile Money Transactions
- Records: 2.17+ Million
- Features: 11
- Database: Microsoft SQL Server 2022

Important Attributes

- Step
- Transaction Type
- Amount
- Sender Account
- Receiver Account
- Original Balance

- Updated Balance
- Fraud Indicator
- Flagged Fraud Indicator

The dataset provides sufficient information to analyze customer behavior and detect suspicious financial activities.

5. Tools & Technologies

The project was implemented using the following technologies:

- Microsoft SQL Server 2022
- SQL Server Management Studio (SSMS)
- SQL
- Common Table Expressions (CTEs)
- Recursive CTEs
- Window Functions
- Aggregate Functions
- CASE Statements
- Joins
- Subqueries

6. Project Methodology

The project follows a structured analytical workflow consisting of multiple stages.

Step 1 – Database Creation

A SQL database was created in Microsoft SQL Server and the PaySim dataset was imported successfully.

Step 2 – Data Validation

The imported data was verified to ensure consistency and completeness. Transaction records, balances, and fraud indicators were examined.

Step 3 – Fraud Detection Analysis

Several SQL queries were developed to detect suspicious transactions, fraudulent customers, high-value transfers, and abnormal account behavior.

Step 4 – Business Analysis

The outputs generated by SQL queries were analyzed to identify fraud trends and provide business recommendations.

7. Database Design

The database consists of a transaction table containing financial transaction records.

Each transaction stores:

- Transaction Type
- Amount
- Sender
- Receiver
- Account Balance Before Transaction
- Account Balance After Transaction
- Fraud Status
- Flagged Fraud Status

This structure enables detailed fraud analysis using SQL.

8. SQL Implementation

The project demonstrates several advanced SQL concepts.

Recursive Fraud Detection

Recursive Common Table Expressions (CTEs) were used to identify connected fraudulent transaction chains across multiple accounts.

Rolling Fraud Detection

Window Functions were used to monitor repeated fraudulent activities performed by the same customer over time.

Multiple CTE Analysis

Multiple Common Table Expressions were combined to identify customers involved in:

- High-value transfers
- No balance changes
- Flagged transactions

Balance Validation

Account balances before and after transactions were validated to detect abnormal financial behavior.

Fraud Investigation

Additional SQL queries were used to identify suspicious customer activity and large transfer patterns.

9. Query Analysis

Several SQL queries were executed during this project.

Query 1

Recursive Fraud Detection

Purpose:

Identify linked fraudulent transactions between multiple accounts.

Query 2

Rolling Fraud Detection

Purpose:

Monitor repeated fraud attempts using Window Functions.

Query 3

Multiple CTE Fraud Detection

Purpose:

Detect high-risk customers using multiple fraud indicators.

Query 4

Balance Validation

Purpose:

Verify transaction consistency by comparing previous and updated balances.

Query 5

High Value Transfer Detection

Purpose:

Identify transactions with unusually large transfer amounts.

10. Results & Findings

The project produced several important findings:

- Transfer transactions showed the highest fraud risk.
- Recursive CTEs successfully detected connected fraud chains.
- Window Functions identified repeated suspicious customer activity.
- Balance validation revealed abnormal transaction behavior.
- Large-value transfers had a greater probability of fraud.
- Multiple suspicious customer accounts were successfully identified using SQL.

11. Business Insights

The analysis provides valuable insights for financial institutions.

- High-value transfer transactions require continuous monitoring.
- Suspicious accounts should be automatically flagged.
- Balance validation improves fraud detection accuracy.
- Window Functions help monitor repeated fraud behavior.
- SQL analytics supports faster fraud investigation and decision-making.

12. Conclusion

This project demonstrates how Microsoft SQL Server can be effectively used for fraud detection and financial analytics. By applying advanced SQL techniques such as Recursive CTEs, Window Functions, and Aggregate Functions, suspicious transaction behavior was successfully identified.

The project highlights the importance of SQL in solving real-world financial problems and demonstrates how data analytics can improve fraud prevention strategies in banking and financial organizations.

13. Future Scope

The project can be further enhanced by incorporating:

- Machine Learning-based Fraud Detection
- Real-time Fraud Monitoring Dashboards
- Power BI Visualization
- Risk Score Prediction Models
- Customer Segmentation
- Cloud Database Integration
- AI-based Fraud Detection Systems

14. References

- I. Microsoft SQL Server Documentation
- II. SQL Server Management Studio Documentation
- III. Microsoft Learn SQL Documentation
- IV. PaySim Mobile Money Transactions Dataset
- V. SQL Window Functions Documentation
- VI. Common Table Expressions (Microsoft Learn)

15. Appendix

Output Screenshots -

a) Database Creation-

Results Messages											
	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
1	1	PAYMENT	9839.64	C1231006815	170136	160296.36	M1979787155	0	0	0	0
2	1	PAYMENT	1864.28	C1666544295	21249	19384.72	M2044282225	0	0	0	0
3	1	TRANSFER	181	C1305486145	181	0	C553264065	0	0	1	0
4	1	CASH_OUT	181	C840083671	181	0	C38997010	21182	0	1	0
5	1	PAYMENT	11668.14	C2048537720	41554	29885.86	M1230701703	0	0	0	0
6	1	PAYMENT	7817.71	C90045638	53860	46042.29	M573487274	0	0	0	0
7	1	PAYMENT	7107.77	C154988899	183195	176087.23	M408069119	0	0	0	0
8	1	PAYMENT	7861.64	C1912850431	176087.23	168225.59	M633326333	0	0	0	0
9	1	PAYMENT	4024.36	C1265012928	2671	0	M1176932104	0	0	0	0
10	1	DEBIT	5337.77	C712410124	41720	36382.23	C195600860	41898	40348.79	0	0
11	1	DEBIT	9644.94	C1900366749	4465	0	C997608398	10845	157982.12	0	0
12	1	PAYMENT	3099.97	C249177573	20771	17671.03	M2096539129	0	0	0	0
13	1	PAYMENT	2560.74	C1648232591	5070	2509.26	M972865270	0	0	0	0
14	1	PAYMENT	11633.76	C1716932897	10127	0	M801569151	0	0	0	0
15	1	PAYMENT	4098.78	C1026483832	503264	499165.22	M1635378213	0	0	0	0
16	1	CASH_OUT	229133.94	C905080434	15325	0	C476402209	5083	51513.44	0	0
17	1	PAYMENT	1563.82	C761750706	450	0	M1731217984	0	0	0	0
18	1	PAYMENT	1157.86	C1237762639	21156	19998.14	M1877062907	0	0	0	0
19	1	PAYMENT	671.64	C2033524545	15123	14451.36	M473053293	0	0	0	0
20	1	TRANSFER	215310.3	C1670993182	705	0	C1100439041	22425	0	0	0
21	1	PAYMENT	1373.43	C20804602	13854	12480.57	M1344519051	0	0	0	0
22	1	DEBIT	9302.79	C1566511282	11299	1996.21	C1973538135	29832	16896.7	0	0
23	1	DEBIT	1065.41	C1959239586	1817	751.59	C515132998	10330	0	0	0
24	1	PAYMENT	3876.41	C504336483	67852	63975.59	M1404932042	0	0	0	0
25	1	TRANSFER	311685.89	C1984094095	10835	0	C932583850	6267	2719172.89	0	0
26	1	PAYMENT	6061.13	C1043358826	443	0	M1558079303	0	0	0	0
27	1	PAYMENT	9478.39	C1671590089	116494	107015.61	M58488213	0	0	0	0
28	1	PAYMENT	8009.09	C1053967012	10968	2958.91	M295304806	0	0	0	0
29	1	PAYMENT	8901.99	C1632497828	2958.91	0	M33419717	0	0	0	0
30	1	PAYMENT	9920.52	C764826684	0	0	M1940055334	0	0	0	0
31	1	PAYMENT	3448.92	C2103763750	0	0	M335107734	0	0	0	0
32	1	PAYMENT	4206.84	C215078753	0	0	M1757317128	0	0	0	0
33	1	PAYMENT	5885.56	C840514538	0	0	M1804441305	0	0	0	0
34	1	PAYMENT	5307.88	C1768242710	0	0	M1971783162	0	0	0	0
35	1	PAYMENT	5031.22	C247113419	0	0	M151442075	0	0	0	0
36	1	PAYMENT	24213.67	C1238616099	0	0	M70695990	0	0	0	0

b) Recursive Fraud Detection -

	initial_account	next_account	step	amount	newbalanceOrig
1	C1305486145	C553264065	1	181	0
2	C1420196421	C972765878	1	2806	0
3	C137533655	C1848415041	1	20128	0
4	C1334405552	C431687661	1	1277212.77	0
5	C1364127192	C1136419747	1	35063.63	0
6	C669700766	C1384210339	1	25071.46	0
7	C1872047468	C116289363	1	235238.66	0
8	C1093223281	C2063275841	2	1096187.24	0
9	C1440057381	C268086000	2	963532.14	0
10	C140702728	C634210724	2	14949.84	0
11	C1375503918	C234430897	2	18627.02	0
12	C1134864869	C118648358	3	10539.37	0
13	C1247938090	C1002031672	3	22877	0
14	C7162498	C945327594	4	10000000	2930418.44
15	C2047521920	C449261773	4	2930418.44	0
16	C540962910	C2127862399	4	169941.73	0
17	C17222024	C410033330	4	13707.11	0
18	C1844941220	C1191544932	4	86070.17	0
19	C1409933277	C162114152	5	120074.73	0
20	C383882703	C1807010830	5	60726.57	0
21	C1775746074	C746009679	5	10119.47	0
22	C2025652936	C1662235713	6	33332.86	0
23	C864622150	C41792607	6	25975.86	0
24	C457596841	C1956477953	6	26768.5	0
25	C903383636	C904300960	6	30521.02	0
26	C666642000	C1315078366	6	216422	0
27	C788790720	C1589547066	6	17320.91	0
28	C1034673425	C1379589736	6	17246	0
29	C777407608	C1790657739	6	10565	0
30	C365589282	C908043384	6	10224	0
31	C558400671	C527482085	6	85354.69	0
32	C1735554279	C12793273	6	13704	0
33	C2101686389	C612008977	7	262434.54	0
34	C1686905633	C824421402	7	21729	0
35	C666654362	C1726301214	7	5460002.91	0
36	C1455969984	C2119910556	7	164	0

c) Rolling Fraud Detection –

	nameOrig	step	fraud_rolling
1	C1002446735	16	1
2	C1003377863	614	1
3	C1004068843	710	1
4	C1004271827	6	1
5	C1005866728	218	1
6	C1011490419	35	1
7	C1011665365	233	1
8	C1011829657	289	1
9	C1012580220	387	1
10	C1013101723	206	1
11	C1014836767	162	1
12	C1015599837	277	1
13	C1016160383	306	1
14	C1016323653	26	1
15	C1016521533	66	1
16	C1018357975	205	1
17	C1018733794	205	1
18	C1019233817	360	1
19	C1020461364	10	1
20	C1022498421	93	1
21	C1025024603	566	1
22	C1026095459	219	1
23	C1027606228	390	1
24	C1029506944	645	1
25	C103076793	229	1
26	C1035541766	279	1
27	C103568596	517	1
28	C1036592129	35	1
29	C1040108506	730	1
30	C1040320190	718	1
31	C1043173812	86	1
32	C1044513187	270	1
33	C104610866	590	1
34	C1046302165	341	1
35	C1046364344	270	1

d) Multiple CTE Fraud Detection-

	nameOrig
1	C689608084
2	C452586515
3	C1035541766
4	C786455622
5	C1892216157
6	C1100582606
7	C19004745
8	C917414431
9	C2140038573
10	C1869569059
11	C992223106
12	C908544136
13	C728984460
14	C193696150

e) Final Result-

	amount	nameOrig	oldbalanceDest	newbalanceDest	new_updated_Balance
1	228863.78	C1202227383	0	228863.78	228863.78
2	225351.95	C465100360	405310.23	630662.18	630662.18
3	188327.21	C1453231433	1680667.69	1868994.9	1868994.9
4	161514.03	C1340382200	358339.53	519853.57	519853.56
5	11269.49	C681564704	35094.83	46364.32	46364.32
6	45367.65	C1805584560	0	45367.65	45367.65
7	294415.54	C954995439	316285.88	610701.41	610701.42
8	29761.71	C70959602	519623.36	549385.07	549385.07
9	69014.92	C2009593531	112275.31	181290.23	181290.23
10	24590.3	C1139696244	118662.54	143252.84	143252.84
11	27074.78	C405738932	51248.53	78323.31	78323.31
12	5320.48	C1767407305	120934.44	126254.93	126254.92
13	88676.2	C1316615146	479864.57	568540.77	568540.77
14	149415.73	C903637641	0	149415.73	149415.73
15	280121.99	C31761217	0	280121.99	280121.99
16	10559.31	C1381142672	2500.58	13059.89	13059.89
17	107289.41	C1930618491	3172.09	110461.5	110461.5
18	177013.1	C1779386593	10910177.51	11087190.62	11087190.61
19	42681.14	C870326862	189812.41	232493.54	232493.55
20	137695.42	C406011546	370895.67	508591.09	508591.09
21	272099.67	C1509557495	0	272099.67	272099.67
22	82996.57	C1053372832	0	82996.57	82996.57
23	3555.86	C31265615	25306.29	28862.15	28862.15
24	7419.22	C2146202775	10050.18	17469.41	17469.4
25	2616.67	C1041162656	11366.25	13982.92	13982.92
26	334808.14	C1884094888	455842.96	790651.1	790651.1
27	17874.87	C1709231096	99544.14	117419.01	117419.01
28	517925.86	C717533794	14193400.79	14711326.65	14711326.65
29	316564.38	C1988011356	464655.51	781219.89	781219.89
30	242837.61	C1313288039	0	242837.61	242837.61
31	573.65	C1711886320	49527.38	50101.03	50101.03
32	30652.8	C1345913047	5843	36495.8	36495.8
33	225496.68	C1032901093	0	225496.68	225496.68
34	226836.63	C371023983	1182230.58	1409067.21	1409067.21
35	309053.88	C1382247810	705697.54	1014751.42	1014751.42
36	34385.77	C469828033	51743	86128.77	86128.77